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Flight Phase Identification and Other Topics Related to Anomaly Detection of Flight Data

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Abstract

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This paper serves as a reference, documenting flight phase identification and other topics related to anomaly detection of flight data. The flight phase identification is needed to preprocess the National Aeronautics and Space Administration (NASA) DASHlink Sample Flight data set. Primarily, the phase as provided in the data set from the Aircraft Condition Management System (ACMS) has some noise which misclassifies flight phase. A Finite State Model (FSM) was developed to condition the ACMS signal to provide eight distinct flight phases: preflight, taxi, takeoff, climb, cruise, approach, and rollout. The algorithm is intended to be included in the data preprocessing step prior to using the data set for machine learning studies. Additionally, this paper provides a reference for linear first-order approximations from first-principles for each phase of flight.

Table of Contents

Abstract ii

List of Acronyms and Acronyms v

Glossary of Key Terms vi

1.0 Introduction 1

1.1 Background 1

1.2 Paper Organization 1

2.0 Flight Phases 3

CICTT Phase 3

Definition 3

Subphase 3

Praxis 3

2.1 Inputs to the Stabilized Flight Phase Architecture 8

2.2 Application of Finite State Machine to Identify Flight Phases 10

2.1.2 Establish Baseline Feature Set 10

3.3.4 Flight Features Importance Based on Flight Dynamics and Control 13

3.0 Other Topics 20

3.1 Rigid-Body Equations and Linearization 20

3.1.1 Non-Linear Equations of Motion in Inertial Frame 20

3.2 Estimating Business Jet Moments of Inertia Using Roskam's Class I Method 25

References 28

List of Acronyms and Acronyms

AI/ML	Artificial Intelligence and Machine Learning
AUROC	Area Under the Receiver Operator Characteristic curve
FDR	Flight Data Recorder
PINN	Physics-informed Neural Network
PR-AUC	Precision-Recall Curve
DHS	Department of Homeland Security
DoD	Department of Defense
SA	Situational Awareness
TSAD	Time-Series Anomaly Detection
NASA	National Aeronautics and Space Administration
FAA	Federal Aviation Administration
IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
U.S.	United States

Glossary of Key Terms

Artificial Neural Network	Distributed computation model that abstracts the data presentation and processing of a human brain as an interconnected set of layers of network nodes called neurons
Flight Data	Information about an aircraft flight that is typically recorded by a flight data recorder. It contains data such as air speed, altitude, heading, engine parameters, control inputs, etc.
Situational Awareness (SA)	Situational awareness is the accurate perception and understanding of all the factors and conditions within the five fundamental risk elements (flight, pilot, aircraft, environment, and type of operation that comprise any given aviation situation) that affect safety before, during, and after the flight (Federal Aviation Association [FAA], 2023).
Physics-informed Learning	Refers to adding constraints to the model learning process based on physical and/or engineering principles
Phases of flight/Flight phases	Referring to the various operational states of aircraft during a flight. These are typically taxi, takeoff, en-route, approach, and landing
Runway Incursion	Presence of unexpected aircraft, vehicle or person in the protected area used for landing and takeoff

1.0 Introduction

1.1 Background

Since the flight dynamics of the aircraft vary significantly depending on the flight phase, the identification is a critical first step of preprocessing before analyzing the flight data. Various methods have been proposed to identify the flight phases. One option is to use the signal from the Aircraft Condition Management System (ACMS). However, it was found that this signal was noisy and misclassified the flight phase in various flights. Instead, this paper defines a Finite State Machine (FSM) and applies two passes of smoothing. The first pass is intended as low-pass filter. The second pass evaluates each time step through the FSM testing state conditions to determine the phase. It was found that certain corner cases needed to be further addressed and consequently a third pass using Viterbi algorithm was used.

Early aircraft analysts simplified first-principle models using linearization and small perturbation techniques. Simplified models provided insights into the primary relationships between various model parameters, particularly for aircraft control systems. In the context of modern machine learning techniques, these simplified models can provide insights into the behavior of the trained models. Additional, these insights could be formalized as physical model constraints.

1.2 Paper Organization

The paper is divided into two sections addressing flight phases and other topics. The flight phases section addresses a method to identify flight phases in the real aircraft

data using a FSM approach. It provides the definition of the flight phases used in the method as well as a discussion on the algorithm.

The second section covers several topics concerning the rigid-body equations of motion. The first topic describes the rigid-body equations of motion and a method to linearize those equations. The second section shows a derivation to estimate the moments of inertia for business jets.

2.0 Flight Phases

The Federal Aviation Administration (FAA) and International Civil Aviation Organization (ICAO) established the Commercial Aviation Safety Team / International Civil Aviation Organization Common Taxonomy Team (CICTT) to standardize the flight phase definitions (CICTT 2013; ICAO 2013). The CICTT standard describes twelve phases of flight. Additionally, it defines five subphases for the enroute phase as shown in

Table 0.1. Phases of Flight Definitions

CICTT Phase	Definition	Subphase	Praxis
Standing (STD)	Prior to pushback or taxi, or after arrival, at the gate, ramp, or parking area, while the aircraft is stationary.		Preflight
Pushback/Towing (PBT)	Aircraft is moving in the gate, ramp, or parking area, assisted by a tow vehicle (tug).		
Taxi (TXI)	The aircraft is moving on the aerodrome surface under its own power prior to takeoff or after landing.		Taxi
Takeoff (TOF)	From the application of takeoff power, through rotation and to an altitude of 35 feet above runway elevation.		Takeoff
Initial Climb (ICL)	From the end of the Takeoff subphase to the first prescribed power reduction, or until reaching 1,000 feet above runway elevation or the VFR pattern, whichever comes first.		Climb
En Route (ENR)	• Instrument Flight Rules (IFR): From completion of Initial Climb through cruise altitude and completion of controlled descent to the Initial Approach Fix (IAF). • Visual Flight Rules (VFR): From completion of Initial Climb through cruise and controlled descent to the VFR pattern altitude or 1,000 feet above runway elevation, whichever comes first.	Climb	
		Cruise	Cruise
		Change of Cruise Level	
		Descent	Approach
		Holding	
Maneuvering (MNV)	Low altitude/aerobatic flight operations.		

Approach (APR)	Instrument Flight Rules (IFR): From the Initial Approach Fix (IAF) to the beginning of the landing flare. Visual Flight Rules (VFR): From the point of VFR pattern entry, or 1,000 feet above the runway elevation, to the beginning of the landing flare.		Approach
Landing (LDG)	From the beginning of the landing flare until aircraft exits the landing runway, comes to a stop on the runway, or when power is applied for takeoff in the case of a touch-and-go landing.		Rollout
Emergency Descent (EMG)	A controlled descent during any airborne phase in response to a perceived emergency situation.		
Uncontrolled Descent (UND)	A descent during any airborne phase in which the aircraft does not sustain controlled flight.		
Post-Impact (PIM)	Any of that portion of the flight which occurs after impact with a person, object, obstacle or terrain.		
Unknown (UNK)	Phase of flight is not discernible from the information available.		Unknown

For this study, several of the CICTT phases have been combined into a single phase. For example, the standing and pushback/towing phases have been combined into a preflight phase. Similarly, this study combines the initial climb and climb phases into a single climb phase. Additionally, the CICTT cruise subphases have been combined into the single cruise phase. It should be noted that the change of cruise level will be referred to as step-up and step-down climbs in this praxis. The CICTT descent and holding subphases and CICTT approach phase have been combined into a single approach phase. The CICTT landing phase is closely mapped to the rollout phase. Rather than trying to detect the landing flare, this study defines the rollout phase from the touchdown point to the subsequent taxi phase. Consequently, the approach phase will conclude with the landing flare prior to rollout. The remaining phase categories of maneuvering,

emergency descent, uncontrolled descent, post-impact if they exist in the data should be detected as flight anomalies.

For time-series analysis, the dataset needs to be segmented into flights to establish temporal sequencing. Because commercial airlines typically follow a routine flight pattern following well-defined phases, the flight data must be broken into phases. Specifically for anomaly detection for commercial airlines, the flight data are pre-segmented based on the established flight phase definitions. This aligns with the fact that for commercial (fixed-wing) aircraft operations change dramatically through the flight but there is a strong similarity across flights for the same phase (Chin et al., 2019; Gavrilovski et al., 2016).

To promote phase segmentation, the NASA DASHlink Sample Flight dataset has a flight phase signal PH 1 hertz signal that is provided by the on-board Aircraft Condition Monitoring System (Matthews, 2012). There are eight phases represented as unknown, preflight, taxi, takeoff, climb, cruise, approach, rollout. The sensor signal had some intermittent noise that misclassified the phase and could potentially cause temporal sequencing issues. One approach is to develop a Finite State Machine (FSM) to determine the aircraft flight phases and transitions directly from aircraft parameters (Soler et al., 2011). This reference followed this approach and developed FSM to resolve noisy flight phase sensor. Figure 1. Architecture For Computing Stabilized Flight Phases shows the states defined along with transition criteria.

Algorithm 1: Two-Pass Flight Phase Smoothing, State Stabilization, and Viterbi Decoding

Input: Master telemetry matrix, global time axis, aircraft type
Output: Finalized stabilized flight phase sidecar array
– Pass 1: Continuous Stream Analysis –
Initialize textsc{FlightPhaseAnalyzer} with aircraft configuration
for each time step in the global telemetry timeline **do**
    Extract altitude, groundspeed, weight-on-wheels, fan speed, and vertical rate
    Query baseline physics models and OpenAP wrappers to compute raw phase confidences
    Assign macro-guard labels (CL, LVL, CR, DE) to downsampled temporal segments
    Segment upsampled telemetry into windows and apply median filtering to smooth out transient noise
end
Save raw computed phases, confidence weights, and macro-guards to master memory
– Pass 2: Stateful Stabilization & Rule Enforcement –
Initialize persistent textsc{FlightPhaseState} tracking object and memory windows
for each telemetry frame in the mission sequence **do**
    Update derived context (rolling movement, thrust envelopes, max altitude apex)
    Evaluate lifecycle flags, touch-and-go resets, and engine shutdown triggers
    Compute independent confidence scores across all 7 discrete flight phases
    if OpenAP macro-guard enforcement is active **then**
        Apply macro-guardrails to shield active climb or cruise states from transient noise
    end
    Apply transition penalties to non-active phases to suppress score chatter
    Evaluate dwell time requirements, hysteresis thresholds, and switch margins
    if best phase meets dominance and dwell criteria **then**
        Update active state tracker, confidence score, and transition timestamps
    else
        Retain current phase and update confidence score
    end
end
– Pass 3: Global Windowed Viterbi Decoding –
Prepare optimized Viterbi payloads using pre-calculated file metadata records
for each sliced temporal mission window **do**
    Construct transition probability matrices and observation emission likelihoods across phases
    Execute the Viterbi decoding algorithm to find the globally optimal sequence of hidden states
    Resolve boundary inconsistencies and penalize improbable phase jumps across chunked boundaries
end
Store finalized globally decoded and stabilized phase array back into master data structures

Figure 1. Architecture For Computing Stabilized Flight Phases

The FSM algorithm evolved into a two-pass system. The first pass up-sampled the input sensor signals using a zero-order hold to 16 Hz and computed a median value over

3 second windows to smooth any sensor noise. The second pass computed the flight phase confidence at each time step based on aircraft state. Additionally, some hysteresis and dwell thresholds were applied to help reduce any noise in the computed phase signal. For nominal flights the flight phase computed by the FSM was more stable than the Aircraft Condition Monitoring System (ACMS) phase. However, for various conditions like climbing or descending to a new cruise altitude, the FSM logic required further refinement.

. Fuzzy logic can be applied to the continuous flights identified by the DBSCAN algorithm to determine the flight phases (Arts, 2021; Sun et al., 2017). The algorithm was codified into an open source Python library called OpenAP (Sun et al., 2020).

The flight phase identification library of OpenAP determines the flight phase using time, altitude, speed and vertical rate inputs to the fuzzy logic algorithm. This does not utilize the other sensors provided by the dataset like weight-on-wheels, landing gear status, and flap positions. Another issue with fuzzy logic algorithms is that they are sensitive to the noise of flight data and result in many transitions (Perrichon et al., 2024). Another limitation of the OpenAP algorithm is it focuses on in-flight operations and only has a single ground phase that does not distinguish the ground phases of preflight, taxi, and rollout. Additionally, OpenAP defines a leveling phase which for this study is considered a subphase of climb, cruise, or descent. Because of these limitations, This algorithm incorporated into the FSM to provide a guard rail for step-up and step-down conditions that establish a new level flight altitude.

To evaluate the phase identification process, the number of phase transitions N_{Trans} and the number of invalid transitions E_{Trans} can be computed. The number of

phase transitions is the sum of the phase differences between two points. The number of invalid transitions is the sum of all phase transitions that violate the state transition matrix. Evaluating the ACMS phase signal and the FSM computed phase signal using these metrics for the first mission for aircraft 652, the FSM computed phase signal reduces the invalid phase transitions from the ACMS phase signal. However, for some flights, the FSM produces invalid phase transitions.

Table 0.2. Three-Pass Phase Stabilization

ACMS Phase Sensor		FSM Computed Phase		Final Stabilized Phase	
N_{Trans}	E_{Trans}	N_{Trans}	E_{Trans}	N_{Trans}	E_{Trans}

To further stabilize the phase, an additional pass was implemented using Hidden Markov Model to define the state transitions and the Viterbi decoder(Perrichon et al., 2024). The three-pass smoothing algorithm was applied to the ACMS flight phase to compute a final stabilized flight phase. The final stabilized flight phase is used as part of the dataset pre-preprocessing pipeline to segment the flight data to train each machine learning model by flight phase.

2.1 Inputs to the Stabilized Flight Phase Architecture

This section describes the inputs to the Finite State Machine (FSM) that is used to compute the stabilized flight phases using the three-pass architecture described in section **Error! Reference source not found.**. The inputs are described in the following table along with usage in the algorithm:

Table 0.3. Finite State Machine Inputs

Sensors	Description	Status	Usage
WOW	Weight-on-Wheels (0=Ground, 1=Airborne)	Active	Ground/Airborne transition
GS	Groundspeed (knots)	Active	Taxi, rollout, and takeoff velocity thresholds
CASS	Selected computed airspeed (knots)	Active	Pilot intent and profile check
BLAC/blac_smooth	Raw longitudinal acceleration and filtered (ft/sec^2)	Active	Inertial movement detection
ALT	Barometric altitude MSL (feet)	Active	Cruise floor and max altitude tracking
ALTR	Altitude rate (ft/min)	Active	Climb and approach verification
RALT	Radar Altimeter AGL (feet)	Active	Terminal approach and landing thresholds
N1C/N1T/N1CO	Engine fan speed current (%RPM), target (%RPM) and compensation flag	Active	Effective thrust, high/low power thresholds
PLA1, PLA2, PLA3, PLA4	Power lever angles for each engine (deg)	Active	Throttle position tracking
SPLG, SPLY	Spoiler deploy green, Spoiler deploy yellow	Active	Spoiler deployment for rollout
ABRK	Airbrake position (deg)	Active	Deceleration confirmation for rollout
FLAP	Flap handle angle (deg)	Active	Takeoff confidence boosting
APFD, VMODE, LMODE	Flight director status, vertical mode, lateral mode	Active	Automation engaged or pilot-in-control
ALTS	Selected altitude MSL (feet)	Active	Pilot intent
acms_phase (PH)	Phase determined by onboard ACMS	Active	
openap_macro	OpenAP phase calculation, 1Hz (CL, CR, LVL, DE)	Active	Phase guard and noise filter
LGDN, LGUP	Landing gear down (=0), Landing gear up (=1)	Inactive	Used for plotting verification of WOW only not in FSM
IVV	Inertial vertical velocity (ft/min)	Inactive	Using ALTR in FSM instead

FADF, FADS	FADEC fail all engines, FADEC status all engines	Inactive	Initial attempt to determine engine status

The third pass uses the Viterbi encoding which defines valid state transitions to enforce valid state transitions and preventing invalid transitions like cruise to rollout without approach. The following table describes the state transitions matrix:

Table 0.4. State Transitions for Viterbi Algorithm

From/To	0: Unknown	1: Preflight	2: Taxi	3: Takeoff	4: Climb	5: Cruise	6: Approach	7: Rollout
0: Unknown	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1: Preflight	Yes	Yes	Yes	No	No	No	No	No
2: Taxi	Yes	No	Yes	Yes	No	No	No	No
3: Takeoff	Yes	No	No	Yes	Yes	No	No	No
4: Climb	Yes	No	No	No	Yes	Yes	No	No
5: Cruise	Yes	No	No	No	No	Yes	Yes	No
6: Approach	Yes	No	No	No	No	No	Yes	Yes
7: Rollout	Yes	No	Yes	No	No	No	No	Yes

2.2 Application of Finite State Machine to Identify Flight Phases

The Finite State Machine was used to identify the flight phases for the National Aeronautics and Space Administration (NASA) DASHlink Sample Flight Data Set(Matthews, 2012). The following sections describe the data set preprocessing pipeline, establishing a baseline feature set and flight features importance based on flight dynamics and control.

2.1.2 Establish Baseline Feature Set

The baseline dataset will be used to train each of the five models by aircraft, flight and by phase such that there will be 35 separate models. The separation by flight is required to maintain consistency in time. Each flight is separated by flight phase as identified by the stabilized flight phases as previously described.

. Once the flights and flight phases are identified, fleetwide statistics are computed so that the dataset can be standardized. One issue that needs to be addressed is handling of the

multi-rate sensor data. Depending on the sensor, the data was recorded in the range of 0.25 hertz to 16 hertz. Consequently, in this study all the data was up-sampled to 16 hertz using a zero-order hold. In other words, if a sensor was recorded at 8 hertz, the array was doubled and neighboring value was repeated to up-sample the signal to 16 hertz. This was done for all 186 sensor parameters as shown in *Figure 2*. The Dataset V1 provides the phase separated, fleet-normalized sensor parameters at 16 hertz.

Although the zero-order hold was chosen to prevent overshooting interpolation issues, the up sampling of the sensor values requires additional smoothing to prevent physically impossible sensor spikes due to step transitions in the data. For this study, the Savitzky-Golay filter was used to smooth the data after up-sampling but also the window sizes of the filter was adjusted based on the ratio of the native sampling rate to the target 16 Hz rate. This technique was applied to all dataset augmentation.

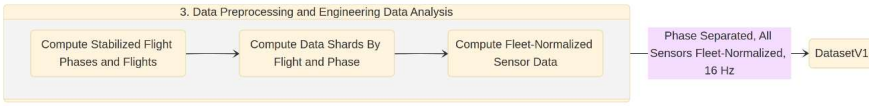


Figure 2. Preprocessing Dataset for Sensors

Based on the linear analysis, the dataset needs to be further augmented with the angular velocities. This is addressed in Dataset V2 as shown in Figure 3. It should be noted that the angular accelerations are also calculated but will be masked and not used as part of the data ingested into the baseline models or physics-informed augmented models.



Figure 3. Preprocessing Dataset for Baseline Models

Because of the need to differentiate to obtain the Euler angle rates, the heading, pitch, and roll needs to smoothed using the Savitzky-Golay filter. This allows the computing of the angular body rates (p, q, r) using the body rate transformation as described in Equation **Error! Reference source not found.**. Another subsequent differentiation computes the angular accelerations (). The fleet-wide statistics per phase are computed to standardize this data and complete the process. The resultant data is stored in a parquet file to supplement the data ingestion process when needed. Again, as mentioned earlier, the accelerations will only be used for the acceleration augmented models.

Because the stabilized phase is being used to segment each flight instead of the Aircraft Condition Monitoring System (ACMS) phase, there appears to be some statistical shifts. For example, for the altitude sensor ALT (MSL feet) that is part of the kinematics group:

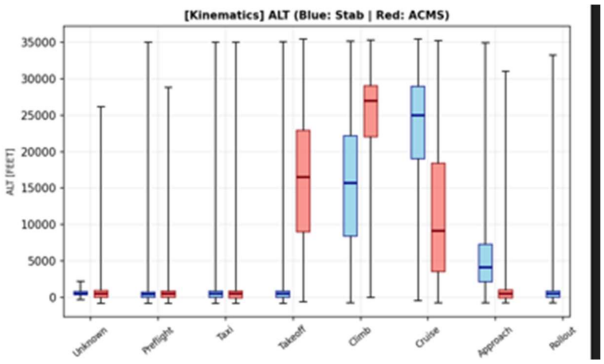


Figure 4. Altitude Sensor Distribution Shift Due to Stabilized Phase

The Figure 4 shows Altitude Sensor distribution using boxplots by phase with the blue boxplots depicting the computed stabilized phase and red depicting the Aircraft

Condition Monitoring System (ACMS) phase from the onboard system. Even for the unclassified Unknown phase, the maximum value appears to be a significant outlier compared to the remaining points. The altitude data in preflight and taxi phases seem to reasonably track each other when comparing the stabilized phase and the ACMS phase. However, there seems to be some issues with the takeoff, climb, cruise, and approach altitudes. The takeoff altitude for the ACMS is too high as the range appears to be more consistent with the climb phase. Similarly, the climb altitude for the ACMS is too high and appears to be more consistent with the cruise phase. Further investigation is warranted to determine if distribution shift is due to an indexing issue with the ACMS data or true data distribution shift due to misclassified ACMS phase data.

Commented [RS1]: Details as shown in Figure 5 and the similar text where you describe the results of your work might go in Chapter 4 if there is space.

3.3.4 Flight Features Importance Based on Flight Dynamics and Control

Simplifying first-principle models using linearization techniques, small angle approximations and perturbation theory to inform preliminary design is an established technique in aircraft design.

Deriving the equations of motion from first principles, the standard 6-DOF nonlinear rigid-body equations of motion can be formulated into three equations governing the translational forces and three equations governing the rotational moment equations:

$$\begin{aligned} \mathrm{F}_X - mg\sin\theta &= m(\dot{u} + qw - rv) \\ \mathrm{F}_Y + mg\sin\phi\cos\theta &= m(\dot{v} + ru - pw) \\ \mathrm{F}_Z + mg\cos\phi\cos\theta &= m(\dot{w} + pv - qu) \\ L &= I_{xx}\dot{p} - I_{xz}\dot{r} + qr(I_{zz} - I_{yy}) - I_{xz}pq \\ M &= I_{yy}\dot{q} + rq(I_{xx} - I_{zz}) + I_{xz}(p^2 - r^2) \\ N &= -I_{xz}\dot{p} + I_{zz}\dot{r} + pq(I_{yy} - I_{xx}) + I_{xz}qr \end{aligned} \tag{1}$$

where:

F_X, F_Y, F_Z	Total external aerodynamic and propulsive forces in the forward, lateral and vertical directions, respectively;
L, M, N	Total external aerodynamic and propulsive moments in the forward, lateral and vertical directions;
u, v, w	Linear velocity components in the body-fixed frame;
p, q, r	Angular velocity components (roll, pitch, yaw rates);
ψ, θ, ϕ	Heading, pitch and roll Euler angles;
$I_{xx}, I_{yy}, I_{zz}, I_{xz}$	Aircraft moments and products of inertia;

The kinematics are further defined by the Body-Rate Transformation given by:

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \tag{2}$$

where $\dot{\phi}, \dot{\theta}, \dot{\phi}$ are the Euler angle rates. Additionally, the velocity of the aircraft in the inertial frame with respect to the body-axis frame:

$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} \tag{3}$$

This formulation can further be simplified by using kinematic and aerodynamic constraints for each phase, applying small-disturbance theory, Taylor series expansion, small angles assumption, and linearizing with control matrix(Etkin & Reid, 1996; Nelson, 1998).

$$\begin{bmatrix} \Delta\dot{u} \\ \Delta\dot{w} \\ \Delta\dot{q} \\ \Delta\dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_w & 0 & -g\cos\theta_0 \\ Z_u & Z_w & U_0 + Z_q & -g\sin\theta_0 \\ M_u + M_{\dot{u}}X_u & M_w + M_{\dot{u}}Z_w & M_q + M_{\dot{u}}(U_0 + Z_q) & -M_{\dot{u}}g\sin\theta_0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta\theta \end{bmatrix} + \begin{bmatrix} X_{\delta_e} & X_{\delta_b} & X_{\delta_T} \\ Z_{\delta_e} & Z_{\delta_b} & Z_{\delta_T} \\ M_{\delta_e} + M_{\dot{u}}X_{\delta_e} & M_{\delta_b} + M_{\dot{u}}X_b & M_{\delta_T} + M_{\dot{u}}X_{\delta_T} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\delta_e \\ \Delta\delta_b \\ \Delta\delta_T \end{bmatrix} \tag{4}$$

$$\begin{bmatrix} \Delta\dot{v} \\ \Delta\dot{p} \\ \Delta\dot{r} \\ \Delta\dot{\phi} \\ \Delta\dot{\psi} \end{bmatrix} = \begin{bmatrix} Y_v & Y_p & Y_r - U_0 & g\,cos\theta_0 & 0 \\ L_v^* & L_p^* & L_r^* & 0 & 0 \\ N_v^* & N_p^* & N_r^* & 0 & 0 \\ 1 & tan\theta_0 & 0 & 0 & 0 \\ 0 & 0 & sec\theta_0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta v \\ \Delta p \\ \Delta r \\ \Delta\phi \\ \Delta\psi \end{bmatrix} + \begin{bmatrix} Y_{\delta_a} & Y_{\delta_r} & Y_{\delta_{steer}} \\ L_{\delta_a}^* & L_{\delta_r}^* & L_{\delta_{steer}}^* \\ N_{\delta_a}^* & N_{\delta_r}^* & N_{\delta_{steer}}^* \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\delta_a \\ \Delta\delta_r \\ \Delta\delta_{steer} \end{bmatrix} \quad (5)$$

$\Delta u, \Delta v, \Delta w$	the perturbations in forward speed, vertical speed, and vertical speed
$\Delta\dot{u}, \Delta\dot{v}, \Delta\dot{w}$	the perturbation velocity rates
θ_o	the reference pitch angle;
g	the acceleration due to gravity
$X_u = \frac{1}{m}\frac{\partial X}{\partial u}, X_w = \frac{1}{m}\frac{\partial X}{\partial w}$, $Z_u = \frac{1}{m}\frac{\partial Z}{\partial u}, Z_w = \frac{1}{m}\frac{\partial Z}{\partial w}$, $Z_q = \frac{1}{m}\frac{\partial Z}{\partial q}$	the stability derivatives normalized by mass
$X_{\delta_e} = \frac{1}{m}\frac{\partial X}{\partial \delta_e}, X_{\delta_b} = \frac{1}{m}\frac{\partial X}{\partial \delta_b}, X_{\delta_T} = \frac{1}{m}\frac{\partial X}{\partial \delta_T}$, $Z_{\delta_e} = \frac{1}{m}\frac{\partial Z}{\partial \delta_e}, Z_{\delta_b} = \frac{1}{m}\frac{\partial Z}{\partial \delta_b}, Z_{\delta_T} = \frac{1}{m}\frac{\partial Z}{\partial \delta_T}$	the stability derivatives normalized by mass with respect to the elevator, brakes, and thrust
$M_u = \frac{1}{I_y}\frac{\partial M}{\partial u}, M_{\dot{u}} = \frac{1}{I_y}\frac{\partial M}{\partial \dot{u}}, M_q = \frac{1}{I_y}\frac{\partial M}{\partial q}$;	the stability derivatives that are normalized by inertia
$M_{\delta_e} = \frac{1}{I_y}\frac{\partial M}{\partial \delta_e}$, $M_{\delta_b} = \frac{1}{I_y}\frac{\partial M}{\partial \delta_b}$, $M_{\delta_T} = \frac{1}{I_y}\frac{\partial M}{\partial \delta_T}$	are the stability derivatives normalized by inertia with respect to the elevator, brakes and thrust
$\Delta p, \Delta q, \Delta r$	the perturbation angular rates
$\Delta\dot{p}, \Delta\dot{q}, \Delta\dot{r}$	the perturbation angular accelerations
$L^* = L + \frac{I_{xz}}{I_x} N$	the cross-coupling derivatives
$N^* = N + \frac{I_{xz}}{I_z} L$	
Y, L, N	are the side-force, rolling and yawing moment derivatives
$\Delta\delta_a, \Delta\delta_r, \Delta\delta_{steer}$	the aileron, rudder and steering inputs

Please see the detailed derivation in Chapter 3 section 3.1.1 Non-Linear Equations of Motion in Inertial Frame.

For the preflight phase, the aircraft is parked and stationary. Using the longitudinal, linear, rigid-body Equation **Error! Reference source not found.**, $U_o = 0$, all aerodynamic stability derivatives are zero

$$\begin{bmatrix} \Delta\dot{u} \\ \Delta\dot{w} \\ \Delta\dot{q} \\ \Delta\dot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & -g \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta\theta \end{bmatrix} + \begin{bmatrix} 0 & X_{\delta_b} & 0 \\ 0 & Z_{\delta_b} & 0 \\ 0 & M_{\delta_b} & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\delta_e \\ \Delta\delta_b \\ \Delta\delta_T \end{bmatrix} \quad (6)$$

Consequently, only the gravity terms remain for the kinematic sensors. Any values in the sensors may indicate sensor biases. Only the braking control terms remain for the configuration sensors.

For the taxi phase, the aircraft is moving slowly on the ground but restricted to the two-dimensional ground plane such climb rate (Δw), the pitch angular velocity (Δq), pitch ($\Delta\theta$) and roll ($\Delta\phi$) are zero. The reference pitch (θ_0) is zero.

$$\begin{bmatrix} \Delta\dot{u} \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & -g \\ 0 & 0 & U_0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} X_{\delta_e} & X_{\delta_b} & X_{\delta_T} \\ 0 & Z_{\delta_b} & 0 \\ 0 & M_{\delta_b} & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\delta_e \\ \Delta\delta_b \\ \Delta\delta_T \end{bmatrix} \quad (7)$$

$$\begin{bmatrix} \Delta\dot{v} \\ \Delta\dot{p} \\ \Delta\dot{r} \\ 0 \\ \Delta\dot{\psi} \end{bmatrix} = \begin{bmatrix} Y_v & Y_p & Y_r - U_0 & g\cos\theta_0 & 0 \\ L_v^* & L_p^* & L_r^* & 0 & 0 \\ N_v^* & N_p^* & N_r^* & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta v \\ \Delta p \\ \Delta r \\ 0 \\ \Delta\psi \end{bmatrix} + \begin{bmatrix} Y_{\delta_a} & Y_{\delta_r} & Y_{\delta_{steer}} \\ L_{\delta_a}^* & L_{\delta_r}^* & L_{\delta_{steer}}^* \\ N_{\delta_a}^* & N_{\delta_r}^* & N_{\delta_{steer}}^* \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\delta_a \\ \Delta\delta_r \\ \Delta\delta_{steer} \end{bmatrix} \quad (8)$$

This basically shows that the most important kinematic sensors are longitudinal velocity like groundspeed, heading, heading rate. The most important control sensors are thrust, directional steering and longitudinal braking. Since our baseline DASHlink dataset is

missing the body-rates and angular accelerations, without the augmentation, lateral steering and lateral braking would not be modeled correctly.

For the takeoff phase, the aircraft starts in ground roll with the brakes released, and the taxi phase equations apply. But at the rotation speed, the nose of the aircraft rotates and lifts off the ground such that the pitch ($\Delta\theta$) and pitch rate (Δq) becomes important. Since the aircraft stays on the runway centerline, the roll ($\Delta\phi$) and yaw ($\Delta\psi$) stay at zero such that only longitudinal force (X) and moment equations (M) and vertical force equation (Z) are important. As the aircraft becomes airborne, the thrust should be near maximum.We can consider the reference pitch (θ_0) to be at the start of takeoff phase while the aircraft is on the ground and is zero.

$$\begin{bmatrix} \Delta\dot{u} \\ \Delta\dot{w} \\ \Delta\dot{q} \\ \Delta\dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_w & 0 & -g \\ Z_u & Z_w & U_0 & 0 \\ M_u + M_{\dot{u}}X_u & M_w + M_{\dot{u}}Z_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta\theta \end{bmatrix} + \begin{bmatrix} X_{\delta_e} & 0 & X_{\delta_T} \\ Z_{\delta_e} & 0 & Z_{\delta_T} \\ M_{\delta_e} + M_{\dot{u}}X_{\delta_e} & 0 & M_{\delta_T} + M_{\dot{u}}X_{\delta_T} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\delta_e \\ 0 \\ \Delta\delta_T \end{bmatrix} \quad (9)$$

This shows that the longitudinal velocity, vertical velocity, pitch rate, pitch, elevator position and engine thrust are important parameters for takeoff. It should be noted that the small-disturbance theory assumed an initial takeoff speed, U_0. Without the augmentation of the pitch angular velocity q, the vertical velocity rate w and the pitch angular acceleration would be underestimated and consequently role of the elevator and engine thrust would be less impactful.

For the climb phase, the flight path angle is positive ($\theta_0 > 0$), high longitudinal velocity (Δu), high thrust ($\Delta\delta_T$). No steering ($\Delta\delta_r = 0$). No braking ($\Delta\delta_b = 0$). The gravity terms are active. Because there is no turning during climb, lateral-directional linear equation is not important.

$$\begin{bmatrix}\Delta\dot{u}\\ \Delta\dot{w}\\ \Delta\dot{q}\\ \Delta\dot{\theta}\end{bmatrix}=\begin{bmatrix}X_u & X_w & 0 & -gcos\theta_0\\ Z_u & Z_w & U_0+Z_q & -gsin\theta_0\\ M_u+M_{\dot{u}}X_u & M_w+M_{\dot{u}}Z_w & M_q+M_{\dot{u}}(U_0+Z_q) & -M_{\dot{u}}gsin\theta_0\\ 0 & 0 & 1 & 0\end{bmatrix}\begin{bmatrix}\Delta u\\ \Delta w\\ \Delta q\\ \Delta\theta\end{bmatrix}+\begin{bmatrix}X_{\delta_e} & 0 & X_{\delta_T}\\ Z_{\delta_e} & 0 & Z_{\delta_T}\\ M_{\delta_e}+M_{\dot{u}}X_e & 0 & M_{\delta_T}+M_{\dot{u}}X_T\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}\Delta\delta_e\\ 0\\ \Delta\delta_T\end{bmatrix}\quad(10)$$

The analysis shows that the longitudinal velocity (Δu), vertical velocity (Δw), pitch angle ($\Delta\theta$), and pitch angular velocity (Δp) are the dominant linear terms. The elevator and thrust sensors are the most important features for the flight dynamics. Again, without the angular rate augmentation, the elevator and thrust dynamics are not properly modeled.

For the cruise phase, the aircraft is in a steady state flight with wings level such that the reference pitch (θ_0), reference roll (ϕ_0) and accelerations are nearly zero.

$$\begin{bmatrix}\Delta\dot{u}\\ \Delta\dot{w}\\ \Delta\dot{q}\\ \Delta\dot{\theta}\end{bmatrix}=\begin{bmatrix}X_u & X_w & 0 & -g\\ Z_u & Z_w & U_0+Z_q & 0\\ M_u+M_{\dot{u}}X_u & M_w+M_{\dot{u}}Z_w & M_q+M_{\dot{u}}(U_0+Z_q) & 0\\ 0 & 0 & 1 & 0\end{bmatrix}\begin{bmatrix}\Delta u\\ \Delta w\\ \Delta q\\ \Delta\theta\end{bmatrix}+\begin{bmatrix}X_{\delta_e} & 0 & X_{\delta_T}\\ Z_{\delta_e} & 0 & Z_{\delta_T}\\ M_{\delta_e}+M_{\dot{u}}X_{\delta_e} & 0 & M_{\delta_T}+M_{\dot{u}}X_{\delta_T}\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}\Delta\delta_e\\ 0\\ \Delta\delta_T\end{bmatrix}\quad(11)$$

$$\begin{bmatrix}\Delta\dot{v}\\ \Delta\dot{p}\\ \Delta\dot{r}\\ \Delta\dot{\phi}\\ \Delta\dot{\psi}\end{bmatrix}=\begin{bmatrix}Y_v & Y_p & Y_r-U_0 & g & 0\\ L_v^* & L_p^* & L_r^* & 0 & 0\\ N_v^* & N_p^* & N_r^* & 0 & 0\\ 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\end{bmatrix}\begin{bmatrix}\Delta v\\ \Delta p\\ \Delta r\\ \Delta\phi\\ \Delta\psi\end{bmatrix}+\begin{bmatrix}Y_{\delta_a} & Y_{\delta_r} & 0\\ L_{\delta_a}^* & L_{\delta_r}^* & 0\\ N_{\delta_a}^* & N_{\delta_r}^* & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}\Delta\delta_a\\ \Delta\delta_r\\ 0\end{bmatrix}\quad(12)$$

The most important longitudinal configuration sensors are elevator and thrust. The most important lateral-directional configuration sensors are ailerons and rudder.

For the approach phase, the flaps are extended and landing gear are deployed so the longitudinal and lateral-directional rigid body equations become:

$$\begin{bmatrix}\Delta\dot{u}\\ \Delta\dot{w}\\ \Delta\dot{q}\\ \Delta\dot{\theta}\end{bmatrix}=\begin{bmatrix}X_u^{flaps} & X_w^{flaps} & 0 & -gcos\theta_o\\ Z_u^{flaps} & Z_w^{flaps} & U_0+Z_q^{flaps} & -\mathrm{gsin}\theta_0\\ M_u^{\mathrm{GE}} & M_w^{\mathrm{GE}} & M_q^{\mathrm{GE}} & -\mathrm{M}_{\dot{u}}gsin\theta_0\\ 0 & 0 & 1 & 0\end{bmatrix}\begin{bmatrix}\Delta u\\ \Delta w\\ \Delta q\\ \Delta\theta\end{bmatrix}+\begin{bmatrix}X_{\delta_e} & 0 & X_{\delta_T}\\ Z_{\delta_e} & 0 & Z_{\delta_T}\\ M_{\delta_e}+M_{\dot{u}}X_{\delta_e} & 0 & M_{\delta_T}+M_{\dot{u}}X_{\delta_T}\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}\Delta\delta_e\\ 0\\ \Delta\delta_T\end{bmatrix}\quad(13)$$

$$\begin{bmatrix} \Delta\dot{v} \\ \Delta\dot{p} \\ \Delta\dot{r} \\ \Delta\dot{\phi} \\ \Delta\dot{\psi} \end{bmatrix} = \begin{bmatrix} Y_v & Y_p & Y_r - U_0 & g & 0 \\ L_v^* & L_p^* & L_r^* & 0 & 0 \\ N_v^* & N_p^* & N_r^* & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta v \\ \Delta p \\ \Delta r \\ \Delta\phi \\ \Delta\psi \end{bmatrix} + \begin{bmatrix} Y_{\delta_a} & Y_{\delta_r} & 0 \\ L_{\delta_a}^* & L_{\delta_r}^* & 0 \\ N_{\delta_a}^* & N_{\delta_r}^* & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\delta_a \\ \Delta\delta_r \\ 0 \end{bmatrix} \quad (14)$$

It shows the most important features are the configuration sensor parameters like the flaps, elevators, thrust, ailerons, and rudder.

3.0 Other Topics

3.1 Rigid-Body Equations and Linearization

This section provides more detailed information about the derivation of the rigid-body equations of motion from first principles and the simplifications involved to highlight the important features by flight phase (Nelson, 1998).

3.1.1 Non-Linear Equations of Motion in Inertial Frame

The following equations apply Newton's second law to model an aircraft translational acceleration and rotational acceleration:

$$\begin{aligned} \vec{F} &= \frac{d}{dt}(m\vec{V}) \\ \vec{M} &= \frac{d}{dt}(\vec{H}) \end{aligned} \tag{15}$$

where F is the external forces acting on the aircraft, m is the mass of the aircraft, V is the velocity of the aircraft, M is the total moments acting on the aircraft and H is the total angular momentum of the aircraft. Assuming constant mass m, constant aircraft inertia tensor I and moving body-axis reference frame,

$$\begin{aligned} \vec{F} &= m\left(\left.\frac{d\vec{V}}{dt}\right|_B + \vec{\Omega} \times \vec{V}\right) \\ \vec{M} &= \left.\frac{d(\vec{I} \cdot \vec{\Omega})}{dt}\right|_B + \vec{\Omega} \times (\vec{I} \cdot \vec{\Omega}) \end{aligned} \tag{16}$$

where $\vec{V} = [u, v, w]^T$is the total velocity vector in the body-axis frame, $\vec{\Omega} = [p, q, r]^T$ is the total angular velocity vector in the body-axis frame, $\vec{F} = [F_X, F_Y, F_Z]^T$ is the total external aerodynamic, thrust and gravity forces acting on the aircraft, $\vec{M} = [L, M, N]^T$ is the external aerodynamic, thrust, and gravity moments acting on the aircraft, $\vec{H} = \vec{I} \cdot \vec{\Omega}$ is

the angular momentum vector and I is the rigid-body inertia tensor. If the xz-plane is assumed to be the plane of symmetry of the aircraft, the products of inertia become $I_{yz} = I_{xy} = 0$. This simplifies the moment equations. The scalar force and moment equations are:

$$\begin{aligned} \mathrm{F}_X - mg\sin\theta &= m(\dot{u} + qw - rv) \\ \mathrm{F}_Y + mg\sin\phi\cos\theta &= m(\dot{v} + ru - pw) \\ \mathrm{F}_Z + mg\cos\phi\cos\theta &= m(\dot{w} + pv - qu) \\ L &= I_{xx}\dot{p} - I_{xz}\dot{r} + qr(I_{zz} - I_{yy}) - I_{xz}pq \\ M &= I_{yy}\dot{q} + rq(I_{xx} - I_{zz}) + I_{xz}(p^2 - r^2) \\ N &= -I_{xz}\dot{p} + I_{zz}\dot{r} + pq(I_{yy} - I_{xx}) + I_{xz}qr \end{aligned} \tag{17}$$

The position and orientation of the aircraft must be related to a fixed frame of reference such that at $t = 0$, the fixed frame and the moving frame coincide. Three consecutive rotations can describe the attitude of the aircraft. The choice of the direction cosine matrix is arbitrary but for aircraft the heading-pitch-roll (3-2-1) is typical. Consequently, the first rotation of heading (ψ) is defined about the z-axis, the second rotation of pitch (θ) about the y-axis, and the third rotation of roll (ϕ) is about the x-axis, the fixed-frame velocity can be related back to the Euler angles and velocity components:

$$\frac{d\vec{R}}{dt} = \vec{R}_{i/b}\vec{V}_B = \vec{R}_1(\phi)\vec{R}_2(\theta)\vec{R_3}(\psi)\vec{V}_B \tag{18}$$

$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} \tag{19}$$

where $\vec{R} = [x, y, z]^T$ is the position vector in the fixed frame and $\vec{V}_B = [u, v, w]^T$ is the velocity vector in the body-axis frame. By integrating this equation with respect to time, the airplane's position in the fixed frame is determined.

Similarly, for the Euler angle vector E in the fixed inertial frame, the direction cosine matrix $\vec{R}_{b/i}$ is the sum of the angular velocities in each direction in the body-axis.

$$\vec{\Omega} = \vec{R}_{b/i}\frac{d\vec{E}}{dt} = \left(\begin{bmatrix}\dot{\phi}\\0\\0\end{bmatrix} + \begin{bmatrix}1 & 0 & 0\\0 & \cos\phi & \sin\phi\\0 & -\sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}0\\\dot{\theta}\\0\end{bmatrix} + \begin{bmatrix}1 & 0 & 0\\0 & \cos\phi & \sin\phi\\0 & -\sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}\cos\theta & 0 & -\sin\theta\\0 & 1 & 0\\\sin\theta & 0 & \cos\theta\end{bmatrix}\begin{bmatrix}0\\0\\\dot{\psi}\end{bmatrix}\right)\frac{d\vec{E}}{dt} \quad (20)$$

$$\begin{bmatrix}p\\q\\r\end{bmatrix} = \begin{bmatrix}1 & 0 & -\sin\theta\\0 & \cos\phi & \sin\phi\cos\theta\\0 & -\sin\phi & \cos\phi\cos\theta\end{bmatrix}\begin{bmatrix}\dot{\phi}\\\dot{\theta}\\\dot{\psi}\end{bmatrix} \quad (21)$$

where $\vec{\Omega} = [p \quad q \quad r]^T$ is the angular velocity vector in body-axis and $\frac{d\vec{E}}{dt} = [\dot{\phi} \quad \dot{\theta} \quad \dot{\psi}]^T$ is the Euler rates in the fixed inertial axis. Inverting the direction cosine matrix and solving for the Euler rates:

$$\begin{bmatrix}\dot{\phi}\\\dot{\theta}\\\dot{\psi}\end{bmatrix} = \begin{bmatrix}1 & \sin\phi\tan\theta & \cos\phi\tan\theta\\0 & \cos\phi & -\sin\phi\\0 & \sin\phi\sec\theta & \cos\phi\sec\theta\end{bmatrix}\begin{bmatrix}p\\q\\r\end{bmatrix} \quad (22)$$

Integrating this expression will compute the Euler rates.

The equations can be linearized using small-disturbance theory where each variable is replaced by a reference value plus a perturbation.

$$\begin{aligned}\vec{V} &= \vec{V}_0 + \Delta\vec{V}\\\vec{\Omega} &= \vec{\Omega}_0 + \Delta\vec{\Omega}\\\vec{F} &= \vec{F}_0 + \Delta\vec{F}\\\vec{\delta} &= \vec{\delta}_0 + \Delta\vec{\delta}\end{aligned} \quad (23)$$

where the velocity perturbation vector is $\Delta\vec{V} = [\Delta u, \Delta v, \Delta w]^T$, the reference velocity vector is $\overrightarrow{V_0} = [U_0, V_0, W_0]^T$, the moment perturbation vector is $\Delta\vec{\Omega} = [\Delta L, \Delta M, \Delta N]^T$, the reference momentum vector is $\overrightarrow{\Omega_0} = [L_0, M_0, N_0]^T$, the force perturbation vector is $\Delta\vec{F} = [\Delta F_X, \Delta F_Y, \Delta F_Z]^T$, the reference momentum vector is $\overrightarrow{F_0} = [F_{X0}, F_{Y0}, F_{Z0}]^T$.

For example, the longitudinal translational and rotational equations, this becomes

$$(\mathrm{F_{X0}} + \Delta\mathrm{F_X}) - \mathrm{mg}\sin(\theta_0 + \Delta\theta) = \mathrm{m}\left(\frac{\mathrm{d(u_0 + \Delta u)}}{\mathrm{dt}} + (\mathrm{q_0 + \Delta q})(\mathrm{w_0 + \Delta w}) - (\mathrm{r_0 + \Delta r})(\mathrm{v_0 + \Delta v})\right) \quad (24)$$

$$(L_0 + \Delta L) = I_{xx}\frac{d(p_0 + \Delta p)}{dt} - I_{xz}\frac{d(r_0 + \Delta r)}{dt} + (q_0 + \Delta q)(r_0 + \Delta r)(I_{zz} - I_{yy}) - I_{xz}(p_0 + \Delta p)(q_0 + \Delta q) \quad (25)$$

Assuming steady, symmetric, straight and level reference trim condition, all time derivatives will be zero ($\frac{du_o}{dt}, \frac{dv_o}{dt}, \cdots = 0$) and angular velocities are zero ($p_0, q_o, r_o = 0$). The trim values for lateral velocity (v_0) and roll angle (ϕ_0) are zero. Aligning the body-axis with longitudinal axis at trim sets the vertical velocity (w_0) to zero. Furthermore, assuming small angle approximations with sine and cosine sum identities:

$$\sin(\theta_0 + \Delta\theta) = \sin\theta_0 \cos\Delta\theta + \cos\theta_0 \sin\Delta\theta \approx \sin\theta_0 + \cos\theta_0 \cdot \Delta\theta$$

$$\cos(\theta_0 + \Delta\theta) = \cos\theta_0 \cos\Delta\theta - \sin\theta_0 \sin\Delta\theta \approx \cos\theta_0 - \sin\theta_0 \cdot \Delta\theta$$

On the left-hand side of the equations, the force and moment changes in the longitudinal directions can be expanded using the Taylor series expansion, keeping only the linear terms and assuming that the longitudinal forces are only a function of u, w, elevator, and thrust and longitudinal moment a function of *v, p, r*, rudder and ailerons then

$$\Delta F_X = \frac{\partial X}{\partial u}\Delta u + \frac{\partial X}{\partial w}\Delta w + \frac{\partial X}{\partial \delta_e}\Delta\delta_e + \frac{\partial X}{\partial T}\Delta\delta_T \quad (26)$$

$$\Delta L = \frac{\partial L}{\partial v}\Delta v + \frac{\partial L}{\partial p}\Delta p + \frac{\partial L}{\partial r}\Delta r + \frac{\partial L}{\partial \delta_r}\Delta\delta_r + \frac{\partial L}{\partial \delta_a}\Delta\delta_a \quad (27)$$

Similarly, this Taylor series expansion can be done for the other force and moment equations. After substituting the force and moment changes equations back into the small-disturbance equations, it turns out that the longitudinal and lateral equations can be decoupled into longitudinal equations and lateral-directional equations. For convenience, these are shown in matrix form:

$$\begin{bmatrix}\Delta\dot{u}\\ \Delta\dot{w}\\ \Delta\dot{q}\\ \Delta\dot{\theta}\end{bmatrix} = \begin{bmatrix} X_u & X_w & 0 & -g\cos\theta_0 \\ Z_u & Z_w & U_0 + Z_q & -g\sin\theta_0 \\ M_u + M_{\dot{u}}X_u & M_w + M_{\dot{u}}Z_w & M_q + M_{\dot{u}}(U_0 + Z_q) & -M_{\dot{u}}g\sin\theta_0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix}\Delta u\\ \Delta w\\ \Delta q\\ \Delta\theta\end{bmatrix} + \begin{bmatrix} X_{\delta_e} & X_{\delta_b} & X_{\delta_T} \\ Z_{\delta_e} & Z_{\delta_b} & Z_{\delta_T} \\ M_{\delta_e} + M_{\dot{u}}X_e & M_{\delta_b} + M_{\dot{u}}X_b & M_{\delta_T} + M_{\dot{u}}X_T \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix}\Delta\delta_e\\ \Delta\delta_b\\ \Delta\delta_T\end{bmatrix} \quad (28)$$

$$\begin{bmatrix}\Delta\dot{v}\\ \Delta\dot{p}\\ \Delta\dot{r}\\ \Delta\dot{\phi}\\ \Delta\dot{\psi}\end{bmatrix} = \begin{bmatrix} Y_v & Y_p & Y_r - U_0 & g\cos\theta_0 & 0 \\ L_v^* & L_p^* & L_r^* & 0 & 0 \\ N_v^* & N_p^* & N_r^* & 0 & 0 \\ 1 & \tan\theta_0 & 0 & 0 & 0 \\ 0 & 0 & \sec\Theta_0 & 0 & 0 \end{bmatrix}\begin{bmatrix}\Delta v\\ \Delta p\\ \Delta r\\ \Delta\phi\\ \Delta\psi\end{bmatrix} + \begin{bmatrix} Y_{\delta_a} & Y_{\delta_r} & Y_{\delta_{steer}} \\ L_{\delta_a}^* & L_{\delta_r}^* & L_{\delta_{steer}}^* \\ N_{\delta_a}^* & N_{\delta_r}^* & N_{\delta_{steer}}^* \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix}\Delta\delta_a\\ \Delta\delta_r\\ \Delta\delta_{steer}\end{bmatrix} \quad (29)$$

where Δu, Δv, Δw is the perturbations in forward speed, vertical speed, and vertical speed; $\Delta\dot{u}$, $\Delta\dot{v}$, $\Delta\dot{w}$ are the perturbation velocity rates; θ_o is the reference pitch angle; g is the acceleration due to gravity; the stability derivatives normalized by mass are $X_u = \frac{1}{m}\frac{\partial X}{\partial u}$, $X_w = \frac{1}{m}\frac{\partial X}{\partial w}$, $Z_u = \frac{1}{m}\frac{\partial Z}{\partial u}$, $Z_w = \frac{1}{m}\frac{\partial Z}{\partial w}$, $Z_q = \frac{1}{m}\frac{\partial Z}{\partial q}$; the stability derivatives normalized by mass with respect to the elevator, brakes, and thrust are $X_{\delta_e} = \frac{1}{m}\frac{\partial X}{\partial \delta_e}$, $X_{\delta_b} = \frac{1}{m}\frac{\partial X}{\partial \delta_b}$, $X_{\delta_T} = \frac{1}{m}\frac{\partial X}{\partial \delta_T}$, $Z_{\delta_e} = \frac{1}{m}\frac{\partial Z}{\partial \delta_e}$, $Z_{\delta_b} = \frac{1}{m}\frac{\partial Z}{\partial \delta_b}$, $Z_{\delta_T} = \frac{1}{m}\frac{\partial Z}{\partial \delta_T}$, the stability derivatives that are normalized by inertia are $M_u = \frac{1}{I_y}\frac{\partial M}{\partial u}$, $M_{\dot{u}} = \frac{1}{I_y}\frac{\partial M}{\partial \dot{u}}$, $M_q = \frac{1}{I_y}\frac{\partial M}{\partial q}$; the stability derivatives normalized by inertia with respect to the elevator, brakes and thrust are $M_{\delta_e} = \frac{1}{I_y}\frac{\partial M}{\partial \delta_e}$, $M_{\delta_b} = \frac{1}{I_y}\frac{\partial M}{\partial \delta_b}$, $M_{\delta_T} = \frac{1}{I_y}\frac{\partial M}{\partial \delta_T}$; $\Delta p, \Delta q, \Delta r$ are the

perturbation angular rates; $\Delta\dot{p}, \Delta\dot{q}, \Delta\dot{r}$ are perturbation angular accelerations; $L^* = L + \frac{I_{xz}}{I_x} N$ and $N^* = N + \frac{I_{xz}}{I_z} L$ are the cross-coupling derivatives; Y, L, N are the side-force, rolling and yawing moment derivatives; $\Delta\delta_a, \Delta\delta_r, \Delta\delta_{steer}$ are the aileron, rudder and steering inputs.

3.2 Estimating Business Jet Moments of Inertia Using Roskam's Class I Method

The moments of inertia using Roskam's Class I method is used for preliminary aircraft design. The following provides the details for estimating the moments of inertia from Roskam's Class I method for a business jet class aircraft (Pegram & Anemaat, 2000).

1. Aircraft Data
    a. Mass (m): 42, 184 kg maximum takeoff weight
    b. Wingspan (b): 26.34 m
    c. Fuselage length (L): 28.55 m
2. Identify Empirical Coefficients
    a. Roll factor ($\bar{R}_x$): 0.25 for wing-mounted jet engines
    b. Pitch factor ($\bar{R}_y$): 0.38 for typical jet transport
    c. Yaw factor ($\bar{R}_z$): 0.39 for transport category structures
3. Compute Roll Inertia

$$I_{xx} = m \cdot (\bar{R}_x \cdot b)^2 = 42184 \cdot (0.25 \cdot 26.34)^2 = 1829192.1\ kg \cdot m^2$$

4. Compute Pitch Inertia

$$I_{yy} = m \cdot \left(\bar{R}_y \cdot L\right)^2 = 42184 \cdot (0.38 \cdot 28.55)^2 = 4965090.6\ kg \cdot m^2$$

5. Compute Yaw Inertia

$$I_{zz} = m \cdot \left(\bar{R}_z \cdot \frac{b+L}{2}\right)^2 = 42184 \cdot (0.39 \cdot \frac{26.34 + 28.55}{2})^2$$

$$= 4832851.4\ kg \cdot m^2$$

3.3 Parameter Groups for DASHlink Sample Flight Data Set

The descriptions of the sensors and a taxonomy to group them is described in the

Table 0.1. DASHlink Sample Flight Data Set Parameter Groups

	Parameter Group	Data Description	Variables
1	Dynamics	Inertial forces and Aerodynamic derivatives and loads	Body Longitudinal Acceleration (BLAC) Cross Track Acceleration (CTAC) Flight Path Acceleration (FPAC) Lateral Acceleration (LATG) Longitudinal Acceleration (LONG) Vertical Acceleration (VRTG)
2	Kinematics	Kinematic State Vectors	Pressure Altitude (ALTG) Altitude Rate (ALTR) Airspeeds (CAS/TAS/MACH) Angle of Attack (AOA1/AOA2/AOAC) Pitch Angle (PTCH) Roll Angle (ROLL) Heading Angle (TH), Magnetic Heading (MH) Track Angles (TRK, TRKM) GPS Position (LATP/LONP) Inertial Vertical Velocity (IVV) Wind Speed (WS) Wind Direction, True (WD) Total Air Temperature (TAT) Static Air Temperature (SAT)
3	Configuration	Configuration & Control Surfaces	Weight On Wheels (WOW) Landing Gear Status (LGDN/LGUP) Squat Switches (MSQT.1/MSQT.2/NSQT) Flap Position (FLAP) Airbrake Position (ABRK) Spoiler Position (SPL.1/SPL.2/SPLG/SPLY) Elevator Position (ELEV.1/ELEV.2)

			Aileron Position (AIL.1/AIL.2) Rudder Position (RUDD) Brake Pressures (BPGR/BPYR) Wind Sheer (WSHR)
4	Propulsion	Engine Information	Actual Fan Speeds (N1.1, N1.2, N1.3, N1.4) Core Speeds (N2.1, N2.2, N2.3, N2.4) Exhaust Gas Temperatures (EGT.1, EGT.2, EGT.3, EGT.4) Fuel Flow (FF.1, FF.2, FF.3, FF.4) Oil Pressure and Temperature (OIP, OIT)
5	Guidance	Flight Management targets. Used for pilot intent.	N1 Command (N1C) N1 Target Setting (N1T) N1 Compensation override flag (N1C0) Selected Altitude (ALTS) Selected Airspeed (CASS/MNS) Selected Heading (HDGS) Selected Vertical Speed (VSPS) Auto-throttle Status (ATEN, A.T) Autopilot Mode Status (APFD) Flight Director Modes (VMODE, LMOD)

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